

Residual-Adaptive Helmholtz-Decomposed PINN for Componentwise Elastic Wavefield Solving in Acoustic Phased-Array Logging With Sparse Data

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Abstract—Accurately solving the multiple components of acoustic wavefields in phased-array logging is significant for wavefield separation and detailed formation characterization. Traditional methods typically rely on elastic wave equations for meshed finite element method (FEM) solutions, which are computationally expensive. Most existing deep learning methods generally lack physical constraints and rely heavily on extensive wavefield data. Furthermore, the inherent coupling properties of the elastic wave equation lead to model failure in solving coupled P- and S-waves without explicit constraints. To address these issues, this work presents a residual-adaptive (RA) physics-informed neural network (RAPINN) for componentwise solution of overlapping multisource forward acoustic wavefields in the phased-array logging scenarios with limited snapshot data. First, a hybrid-driven framework integrating prior physical knowledge with sparse data constraints is proposed. It embeds the governing equations describing wavefield propagation behavior characteristics into the loss function. By training with sparse data, wavefield prediction accuracy is enhanced while the required wavefield data is significantly reduced. Second, the coupled elastic wave equations are decomposed via Helmholtz decomposition to derive potential-based wave equations, enhancing the explicit constraints for P- and S-waves. Third, a pointwise weight assignment strategy based on a residual adaptive mechanism is proposed. The pointwise weights of the sampling points are adapted according to their residual gradients, thereby enhancing the focus on details in difficult-to-learn regions. Finally, exploratory attempts are made to investigate the fluid–solid structure of the borehole, establishing a multisource coupled elastic wavefield surrogate model based on RAPINN under limited data. Numerical examples conducted under different source quantities, synthetic wavefield deflection angles, boundary conditions (BCs), and media demonstrate that the RAPINN provides a feasible reference for acoustic wavefield solving problems.

Index Terms—Acoustic logging, elastic wave equation, Helmholtz decomposition, P- and S-waves solution, residual-adaptive (RA) physics-informed neural networks (RAPINNs).

I. INTRODUCTION

ACOUSTIC phased-array-based logging technologies are widely used to detect geological anomalies ahead of the

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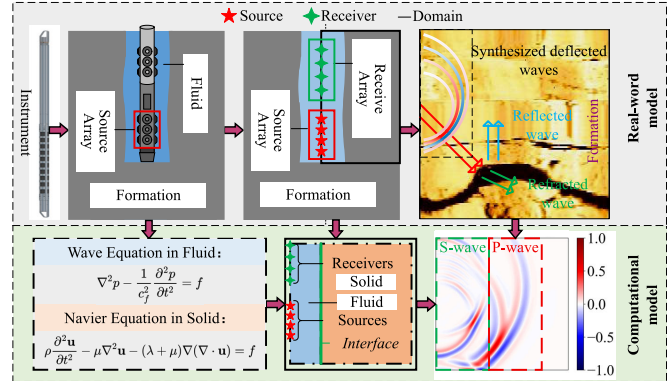


Fig. 1. Schematic of phased-array logging and computational model.

drill bit, playing a vital role in real-time geological exploration and deep formation oil reservoir detailed evaluation [1], [2]. As shown in Fig. 1, during logging, the phased-array ultrasonic transducers excite a deflected acoustic wavefield in the medium. Especially in the fluid medium, it propagates as a longitudinal wave (P-wave), and in the solid medium, it propagates as a coupled longitudinal and transverse wave (P- and S-waves). When encountering geological bodies or interfaces of different media, the different acoustic impedance of the medium causes the acoustic waves to reflect and refract at the interfaces [3]. In this case, the interface medium determines the characteristics of the reflected waves that contain information about the geological body ahead. Therefore, accurately understanding and solving the propagation characteristics of the acoustic phased-array-based forward radiation wavefield in the medium is important for acoustic phased-array-based logging technology [4].

Based on the widely established 2-D plane strain assumption [6] in seismic wavefield simulation [7], [8], the acoustic wavefield solution problem can be simplified to a 2-D model. This simplification leverages the specific axisymmetric structure of boreholes, allowing primary attention toward axial (y) and radial (x) wavefield propagation characteristics. Nevertheless, this simplification neither reduces the inherent complexity of phased-array logging scenarios [9], [10]. As shown in Fig. 1, within the borehole structure, the internal drilling fluid is typically a fluid medium, while the formation is a solid. Their differing properties determine distinct acoustic wavefield characteristics. Simultaneously, due to the spatial and temporal overlaps of the multisource acoustic wavefield generated by phased-array excitation, different mode waves, such as P- and

S-waves, are mutually coupled. Therefore, accurately solving the evolution of the forward radiation wavefield in the acoustic phased-array-based logging scenario is challenging.

The dynamic evolution of the acoustic wavefield is described by elastic wave equations [11]. Therefore, the core task of solving the elastic wavefield involves solving the elastic wave equations. However, due to the high complexity of the multiple equations and the tremendous difference in medium properties, it is difficult to find analytical solutions for elastic wave equations. This has also led numerical solution methods that rely heavily on mesh discretization, such as the finite element method (FEM) [12], [13] and finite difference method (FDM) [14], dominating the field in recent decades. By approximating solutions within discrete mesh cells, these methods ultimately obtain a global approximate solution. For example, Yang [15], [16] developed a nearly analytical discrete method. By incorporating wavefield gradient information and higher-order interpolation approximations, the solution process for seismic wavefield problems is deeply optimized, thereby significantly improving computational efficiency with guaranteed accuracy. It is worth noting that the grid discretization strategy significantly impacts the computational results. Coarse grids typically yield lower computational accuracy, while fine grids substantially increase computational costs for hardware [17].

With the advancement of deep learning, physics-informed neural networks (PINNs) are attracting more attention as a representative framework for solving partial differential equations. They integrate partial difference equations (PDEs), boundary conditions (BCs), and initial conditions (ICs) into the loss function to guide the training process of deep neural networks (DNNs) [18]. This strategy approximates the solution to PDEs by forcing the neural network's output to minimize a loss function comprising PDEs, BCs, and ICs. Compared to purely data-driven approaches, PINNs significantly reduce the training data cost while increasing physical prior knowledge. Consequently, PINNs have applications in many fields, such as mechanics [19], [20], [21], acoustics [22], [23], [24], [25], and geophysics [26], [27], [28].

In PINN-based seismic wavefield modeling, many inspiring works have been reported [29], [30]. Lu et al. [31] proposed a PIRNN framework for solving the vertical seismic profiling full waveform inversion problem. PIRNN achieves excellent seismic waveform inversion results by integrating prior physical knowledge with sparse data constraints. Furthermore, Ren et al. [32] employed domain decomposition techniques to establish a framework for solving semi-infinite seismic wavefields in a multilayer medium using PINN. Rao et al. [33] proposed a PINN with mixed-variable outputs to model elastic dynamics without requiring labeled data. This PINN framework effectively models wave propagation in static and dynamic problems, as well as in truncated domains. Regarding mode separation, Mu et al. [34] proposed a separationPINN to force the DNN to output the P- and S-wave wavefields independently by explicitly constraining the P- and S-wave components. Furthermore, a deep numerical method (DeepNM) was designed to solve the 1-D wave equation [35]. By integrating traditional numerical methods

with deep learning techniques, DeepNM achieves a significant boost in computational speed with comparable accuracy. Similarly, Chang et al. [36] further developed the deep Lax-Wendroff correction (DeepLWC) method for solving 2-D hyperbolic wave equations. DeepLWC significantly improves computational speed and reduces memory requirements. These existing works confirm that PINNs accurately model the propagation process of seismic wavefields based on elastic wave equations. However, the forward radiation wavefield exhibits high complexity in acoustic phased-array logging scenarios. The synthetic wavefields excited by different sources exhibit spatial-temporal coupling. Furthermore, the unique structure of the borehole induces mutual conversion between different mode waves, while also causing acoustic reflection and refraction phenomena. These phenomena present a highly challenging task for numerically simulating acoustic phased-array-based forward radiation wavefields using PINNs.

On the other hand, PINNs typically depend on empirically fixed weights during training. This approach results in imbalanced convergence speeds among different loss terms, hindering the model's accurate convergence [37], [38]. These issues potentially affect the integrity of the wavefield reconstruction and consequently reduce the accuracy of solving overlapping wavefields. To address this issue, McClenny and Braga-Neto [39] introduced the pointwise adaptive weighting method into PINNs to enhance performance. However, it is important to note that the complex gradient computation significantly increases training time. Furthermore, standard PINN struggles to solve coupled P- and S-waves simultaneously due to the absence of explicit constraints for different mode waves. These issues have prompted us to reexamine the way elastic wave equations are embedded into the loss function and the optimization methods for the pointwise weights adaptive mechanism.

To achieve an efficient and accurate evolutionary solution for 2-D overlapped wavefields in acoustic phased-array logging scenarios, this work explores the feasibility of modeling propagation, reflection, and refraction phenomena of forward-propagating acoustic wavefields employing improved PINN under limited snapshot data constraints. The main contributions are as follows.

- 1) A hybrid-driven framework, combining physical priors and sparse data constraints, is proposed to construct an acoustic phased-array-based forward-radiation wavefield surrogate model. Like PINNs, the framework integrates the governing equations describing acoustic wavefield propagation characteristics into the loss function, guiding the training process jointly with sparse data. This approach not only enhances the physical interpretability and wavefield prediction quality but also significantly reduces the required training wavefield datasets.
- 2) The Helmholtz decomposition is employed to decompose the elastic wave equation without explicit constraints into potential-based wave equations with independent constraints for P- and S-waves. These explicit constraints achieve explicit P- and S-wave decoupling. Further integration with the vector decomposition enables the componentwise solution of multiple

wavefield variables, including displacement, velocity, stress, and strain.

- 3) A residual-adaptive (RA) mechanism is designed and integrated into the standard PINN framework (RAPINN). The RA mechanism addresses the overfitting tendency caused by fixed loss term weights by applying trainable weights at each training point. Its optimization direction synchronized with network weights, thereby refining residual point weight distribution. The RA mechanism achieves precise reconstruction of overlapping wavefields without increasing training costs.
- 4) Given the unique structure of the borehole, this work employs RAPINN tentatively to develop a surrogate model of the phased-array-based forward acoustic wavefield. Under sparse data supervision and continuity conditions at the interface, this method efficiently solves fluid–solid coupling acoustic wavefield problems in the phased-array logging scenarios.

Section II provides a comprehensive overview of RAPINN, detailing its design and implementation. Section III illustrates the effectiveness of RAPINN for componentwise solving time-domain elastic wave propagation problems under different conditions, including multiple sources, multiple BCs, and different deflection angles. Examples are given to demonstrate the performance and versatility of the RAPINN. Finally, conclusions are drawn in Section IV.

II. METHODOLOGY

A. Elastic Wave Equations and Helmholtz Decomposition

1) Elastic Wave Equations in Fluid and Solid Media:

Since fluids only transmit compressive stress, P-waves are the only type of wave that exists in the fluid medium. Under the assumption of ignoring the axial information, the homogeneous wave equation for a 2-D acoustic medium under the condition of constant density has the form

$$\nabla^2 p - \frac{1}{c_f^2} \frac{\partial^2 p}{\partial t^2} = f \quad (1)$$

where c_f is the P-wave velocity of the fluid. p is the solved pressure wavefield, which is a function of the spatial–temporal location $\mathbf{X} = \{x, y, t\}$. The source term $f(x, y, t)$ determines the ICs. In this work, the ICs are given by two early time steps of the wavefield. Therefore, the source term $f(x, y, t)$ can be set to zero, i.e., $f \equiv 0$.

In contrast to fluids, elastic solid media not only transmit P-waves but also possess shear elasticity, enabling them to propagate both P- and S-waves simultaneously. This phenomenon also introduces a greater degree of complexity to the acoustic wavefield within the context of the elastic medium. The specifically governing equations of interest are given by [40]

$$\rho \frac{\partial^2 \mathbf{u}}{\partial t^2} = \mu \nabla^2 \mathbf{u} + (\lambda + \mu) \nabla (\nabla \cdot \mathbf{u}) \quad (2)$$

where ρ is the density, and λ and μ represent the Lamé constants of the medium. $\mathbf{u} = \{u_x, u_y\}^T$ represent the displacement vectors. Equation (2) describes the acoustic wavefield with displacement as the primary variable. Based on (2), velocity,

stress, and strain are further solved through the obtained displacement vectors, thereby enabling the acoustic wavefield to be described precisely. Equation (3) lists the relationships between displacement, velocity, strain, and stress

$$\begin{aligned} \mathbf{v} &= \frac{\partial \mathbf{u}}{\partial t} \\ \boldsymbol{\epsilon} &= \frac{1}{2} [(\nabla \mathbf{u})^T + \nabla \mathbf{u}] \\ \boldsymbol{\sigma} &= \mathbf{C} : \boldsymbol{\epsilon} \end{aligned} \quad (3)$$

where $\mathbf{v} = \{v_x, v_y\}^T$, $\boldsymbol{\epsilon} = \{\epsilon_{xx}, \epsilon_{xy}, \epsilon_{yy}\}^T$, and $\boldsymbol{\sigma} = \{\sigma_{xx}, \sigma_{xy}, \sigma_{yy}\}^T$ are the velocity, strain, and stress vectors, respectively. $\mathbf{C}$ represents the constitutive tensor. Note that $\{\sigma_{xx}, \sigma_{yy}\}$ means the normal stresses along the x - and y -dimensions, respectively; σ_{xy} is the shear stress. In essence, (3) represents the governing equations of elastic dynamics under the plane strain assumption [6], which is also the foundation for simplifying 2-D problems. Furthermore, the free boundary at the bottom is used to generate the reflected wavefield. Similarly, the fluid–solid interface is defined by the velocity continuity and stress balance conditions. These BCs are defined as follows:

$$\begin{aligned} \boldsymbol{\sigma} \cdot \mathbf{n}_b &= 0 \\ \mathbf{v}_f \cdot \mathbf{n}_{it} &= \mathbf{v} \cdot \mathbf{n}_{it} \\ -p &= \boldsymbol{\sigma} \cdot \mathbf{n}_{it} \end{aligned} \quad (4)$$

where $\mathbf{n}_b$ and $\mathbf{n}_{it}$ are the norm vector of the bottom (x -direction) reflected boundary and fluid–solid interface along the axial direction (y -direction), respectively. $\mathbf{v}$, $\boldsymbol{\sigma}$, p , and $\mathbf{v}_f$ represent the velocity and stress in the solid medium, and pressure and velocity in the fluid medium. As mentioned above, (4) considers two typical cases. The first subformula represents the free BC, where the normal stress is zero. It simulates the bottom boundary to generate reflected waves. The second and third subformulas, respectively, represent the velocity continuity and pressure balance conditions at the fluid–solid interface along the axial direction. They characterize the propagation behavior of the wavefield on the borehole structure. These BCs independently constrain the propagation behavior at different property boundaries and together describe the complex reflection behavior of the wavefield. On the other hand, it is worth noting (2) describes the propagation behavior of two coupled-mode waves. This also explains why it fails to characterize the independence of the two-mode waves inherently. In other words, it does not impose an explicit, independent constraint for the two-mode waves, but rather a coupled, implicit constraint. For PINNs, implicit constraints are difficult to optimize and are unable to learn different coupled-mode waves. Therefore, (2) must be decomposed into a form with independent explicit constraints for P- and S-waves.

2) *Helmholtz Decomposition:* In an elastic medium, the wavefield $\mathbf{u}$ contains P-wave displacement and S-wave displacement components. Based on the Helmholtz decomposition [41], [42], the wavefield $\mathbf{u}$ can be decomposed into a scalar potential ϕ and a vector potential $\boldsymbol{\psi}$

$$\mathbf{u} = \nabla \phi + \nabla \times \boldsymbol{\psi} \quad (5)$$

where $\nabla\phi$ and $\nabla \times \psi$ represent the P- and S-wave displacements, respectively. These fields must satisfy the following requirements: P-waves are curl-free, and S-waves are divergence-free, as shown in the following equation:

$$\begin{aligned}\nabla \times \nabla\phi &= 0 \\ \nabla \cdot \nabla \times \psi &= 0 \\ \nabla \cdot \psi &= 0.\end{aligned}\quad (6)$$

Therefore, the coupled wavefield displacement $\mathbf{u}$ is decomposed into two independent P- and S-waves potential function components, thereby deriving explicit constraints on wavefield behavior. This is significant for PINNs' learning of coupled wavefield features. Thus, taking (5) into (2) to obtain the governing equation with Helmholtz decomposition shown in the following equation:

$$\nabla \left[(\lambda + 2\mu) \nabla^2 \phi - \rho \frac{\partial^2 \phi}{\partial t^2} \right] - \nabla \times \left(\mu \nabla^2 \psi - \rho \frac{\partial^2 \psi}{\partial t^2} \right) = 0. \quad (7)$$

Under the assumption of ignoring azimuth information, (2) is further simplified by the potential function and Helmholtz decomposition. This derives the governing equations in (8), serving as independent constraints for P- and S-waves

$$\begin{aligned}\nabla^2 \phi - \frac{1}{c_p^2} \frac{\partial^2 \phi}{\partial t^2} &= 0 \\ \nabla^2 \psi - \frac{1}{c_s^2} \frac{\partial^2 \psi}{\partial t^2} &= 0\end{aligned}\quad (8)$$

where, since $\psi_x = \psi_y = 0$, $\psi = \psi_z$. The c_p and c_s represent the P- and S-wave velocities, respectively. It is evident that transforming the coupled (2) into two independent P- and S-waves constraint equations through Helmholtz decomposition enhances the constraint capability for P- and S-waves. This strategy ensures the coupled wavefield governing equations in solids are consistent in form with the wave equation in fluids, greatly reducing the complexity of physical constraints and the training difficulty. The displacement is then expressed in terms of the scalar potential ϕ and vector potential ψ . Furthermore, (3) is used to derive multiple physical variables, including velocity, stress, and strain. Finally, to more precisely describe the dynamic response of the coupled wavefields, the componentwise decomposition scheme is used in the x - and y -directions within a Cartesian coordinate system to analyze each component.

B. Physics-Informed Neural Network

PINNs integrate PDEs into the loss function as the regularization terms to guide the training process. This incorporation mechanism aims to enhance interpretability, distinguishing it from traditional data-driven deep learning approaches. For the acoustic equations and BCs in the phased-array logging scenarios mentioned in Section II-A, the residual loss terms are constructed and embedded into the loss function. They are defined in the following equation:

$$\mathcal{L}_r = \frac{1}{N_r} \sum_{i=1}^{N_r} \left(\left\| \nabla_i^2 \hat{p}_i - \frac{1}{c_f^2} \frac{\partial^2 \hat{p}_i}{\partial t_i^2} \right\|_2^2 + \left\| \nabla_i^2 \hat{\psi}_i - \frac{1}{c_s^2} \frac{\partial^2 \hat{\psi}_i}{\partial t_i^2} \right\|_2^2 \right)$$

$$\begin{aligned}& + \left\| \nabla_i^2 \hat{\phi}_i - \frac{1}{c_p^2} \frac{\partial^2 \hat{\phi}_i}{\partial t_i^2} \right\|_2^2 \\ \mathcal{L}_{bc} &= \frac{1}{N_{bc}} \sum_{i=1}^{N_{bc}} \|\hat{\sigma}_i \cdot \mathbf{n}_b\|_2^2 \\ \mathcal{L}_{it} &= \frac{1}{N_{it}} \sum_{i=1}^{N_{it}} \left(\|\hat{\mathbf{v}}_{fi} \cdot \mathbf{n}_{it} - \hat{\mathbf{v}}_i \cdot \mathbf{n}_{it}\|_2^2 + \|\hat{p}_i - \hat{\sigma}_i \cdot \mathbf{n}_{it}\|_2^2 \right)\end{aligned}\quad (9)$$

where $\mathcal{L}_r$, $\mathcal{L}_{bc}$, and $\mathcal{L}_{it}$ represent the PDEs loss, BCs loss, and interface loss, respectively. The subscript $*_i$ and superscript $\hat{*}$ are the i th sampling point and predicted variables. N_r , N_{bc} , and N_{it} represent the number of sampling points. $\|\cdot\|_2$ is the $\mathcal{L}_2$ norm. As described in Section II-A, $\mathcal{L}_{bc}$ and $\mathcal{L}_{it}$ impose constraints on the bottom reflected boundary and the fluid–solid interface, respectively. On the other hand, the initial wavefields are given by the two early snapshots. They determine the propagation direction and velocity of the incident wavefields. Furthermore, two additional random intermediate-time step wavefield snapshots (including displacement and stress) are introduced as supervisory constraints to enhance the modeling accuracy of the overlapping acoustic wavefield. These supervisory constraints are collectively termed data loss, which includes initial wavefield pointwise data loss ($\mathcal{L}_{ic}$) and snapshot data loss items ($\mathcal{L}_{sn}$) at any given moment. The data loss is defined as follows:

$$\begin{aligned}\mathcal{L}_{data} &= \mathcal{L}_{ic} + \mathcal{L}_{sn} \\ &= \frac{1}{N_{ic}} \sum_{i=1}^{N_{ic}} \|\mathbf{U}_i^{ic} - \hat{\mathbf{U}}_i^{ic}\|_2^2 + \frac{1}{N_{sn}} \sum_{i=1}^{N_{sn}} \|\mathbf{U}_i^{sn} - \hat{\mathbf{U}}_i^{sn}\|_2^2.\end{aligned}\quad (10)$$

The $\mathbf{U}$ and $\hat{\mathbf{U}}$ represent the FEM reference value and prediction, respectively. N_{ic} and N_{sn} are the number of sampling points. It is worth emphasizing that data loss requires only a limited snapshot dataset—just two early snapshots and two intermediate snapshots, effectively constraining the data from four time points. Therefore, data loss also represents a sparse data constraint.

Furthermore, these residuals further form the objective function $\mathcal{L}$ in (11), which serves as the objective function for training

$$\mathcal{L} = \lambda_r \mathcal{L}_r + \lambda_{bc} \mathcal{L}_{bc} + \lambda_{data} \mathcal{L}_{data} + \lambda_{it} \mathcal{L}_{it}. \quad (11)$$

The λ_r , λ_{bc} , λ_{data} , and λ_{it} represent the weights assigned to each loss term. These weights are manually adjusted based on empirical evidence. The loss function incorporates physical prior knowledge and sparse known data, enabling a hybrid physics- and data-driven approach. This approach reduces heavy dependence on numerical wavefield solutions while ensuring reasonable accuracy outside the training datasets. The loss function is minimized to update the network parameters to approximate the solution of the PDEs. Once the training is complete, the trained RAPINN is able to continuously predict phased-array wavefields and perform waveform analysis. This provides an efficient method for predicting wavefields.

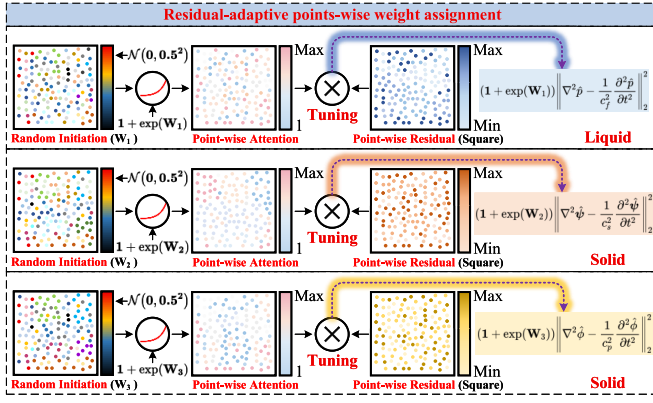


Fig. 2. Schematic of the RA pointwise weight assignment.

C. RA Pointwise Weight Assignment

As described in Section II-B, a fixed weight (usually 1) is applied to each sampling point to directly compute the loss in the standard PINN. This prevents loss at the different sampling points to update based on their own gradients, which may lead to issues such as gradient spatial imbalance and local overfitting. To address this issue, a residual adaptive mechanism is developed, which adaptively updates pointwise weights based on the loss values of each PDE. This enables precise learning in difficult-to-learn regions. As shown in Fig. 2, a Gaussian matrix is randomly initialized for each governing equation, and an exponential operation is performed with a bias to obtain pointwise weights corresponding to the sampling points. The weight matrix is then multiplied by the loss value to fine-tune each PDE loss. Consequently, the PDEs residual loss term $\mathcal{L}_r$ is expressed as follows:

$$\mathcal{L}_r(\mathbf{W}) = \frac{1}{N_r} \sum_{i=1}^{N_r} \left[(1 + \exp(\mathbf{W}_{1i})) \left\| \nabla_i^2 \hat{p}_i - \frac{1}{c_f^2} \frac{\partial^2 \hat{p}_i}{\partial t_i^2} \right\|_2^2 + (1 + \exp(\mathbf{W}_{2i})) \left\| \nabla_i^2 \hat{\psi}_i - \frac{1}{c_s^2} \frac{\partial^2 \hat{\psi}_i}{\partial t_i^2} \right\|_2^2 + (1 + \exp(\mathbf{W}_{3i})) \left\| \nabla_i^2 \hat{\phi}_i - \frac{1}{c_p^2} \frac{\partial^2 \hat{\phi}_i}{\partial t_i^2} \right\|_2^2 \right] \quad (12)$$

where $\mathbf{W} = \{\mathbf{W}_1, \mathbf{W}_2, \mathbf{W}_3\}$ is the spatial weight matrix, and its range is $[1, +\infty]$. Specifically, the gradient for $\mathbf{W}_{3i}$ at i th sampling point is

$$\frac{\partial \mathcal{L}_{r_i}}{\partial \mathbf{W}_{3i}} = \frac{\exp(\mathbf{W}_{3i})}{N_r} \left\| \nabla_i^2 \hat{\phi}_i - \frac{1}{c_p^2} \frac{\partial^2 \hat{\phi}_i}{\partial t_i^2} \right\|_2^2 \quad (13)$$

which is controlled by different PDEs. This method maintains a consistent optimization direction between the weight matrix and network parameters. Therefore, this approach saves training time while ensuring the weight matrix adaptive updates based on its own gradients. Meanwhile, since the weight $\mathbf{W} > 1$, it does not degenerate into the standard PINN. The objective function is rewritten as follows:

$$\min_{\theta, \mathbf{W}} \mathcal{L}(\mathbf{W}, \theta) = \lambda_r \mathcal{L}_r(\mathbf{W}, \theta) + \lambda_{bc} \mathcal{L}_{bc}(\theta)$$

$$+ \lambda_{data} \mathcal{L}_{data}(\theta) + \lambda_{it} \mathcal{L}_{it}(\theta) \quad (14)$$

where $\mathbf{W} = \{\mathbf{W}_1, \mathbf{W}_2, \mathbf{W}_3\}$. Within the loss function, the weights $\mathbf{W}$ and parameters θ are mutually coupled and evolve synchronously, thereby forming a dynamic mechanism that standard PINN cannot replicate. In other words, the introduction of the weight $\mathbf{W}$ changes the optimization dynamics.

D. RAPINN-Based Solution Framework

As described in Sections II-A and II-B, the proposed RAPINN simulates the evolution of acoustic wavefields by solving the elastic wave equations based on Helmholtz decomposition. This allows phased-array wavefields to be modeled in both homogeneous solid medium and fluid–solid composite models under sparse data conditions. Fig. 3 illustrates the workflow for the fluid–solid coupled wavefields modeling case using RAPINN. First, the traditional FEM is used to solve a simplified 2-D acoustic phased-array wavefield. The resulting wavefield snapshot serves as the reference benchmark. Then, two early snapshots are used as the ICs to give the initial wavefield energy. Second, RAPINN takes discrete spatial–temporal sampling points $\mathbf{X}$ as input and outputs pressure p , scalar potential ϕ , and vector potential ψ . Generally, at each collocation point, the strong form governing equation in (1) and (8) should be satisfied. The output pressure p , scalar potential ϕ , and vector potential ψ are expressed as

$$\hat{p}, \hat{\phi}, \hat{\psi} = \mathcal{F}(\mathbf{X}|\theta) \quad (15)$$

where $\mathbf{X} = \{x, y, t\} \in \mathbb{R}^3$ refers to the discrete spatial–temporal sampling points. θ is the trainable parameter of the neural network that performs the mapping function $\mathcal{F}$. Let $\mathcal{F}(\mathbf{X}|\theta): \mathbb{R}^{d_x} \rightarrow \mathbb{R}^{d_y}$ be an L -layer fully connected network with input vector $\mathbf{X}$, output vector $\mathbf{Y}$, and network parameters θ . It follows nested transformations:

$$\mathbf{z}^l = \text{Tanh}^l(\theta^l \odot \mathbf{X}), \quad l = 1, \dots, L-1 \quad (16)$$

where $\mathbf{z}^0 \equiv \mathbf{X}$ and $\mathbf{z}^L \equiv \mathbf{Y}$ are inputs and outputs of the model layer, and θ^l is parameters of each layer l , includes weights and biases. Operation $\odot$ represents the computational process of network parameters and input data. The functions Tanh are called hyperbolic tangent activation functions and make the network nonlinear with respect to the inputs. For instance, a mapping $\mathcal{F}$ functional of scalar potential ϕ , with three hidden layers and with activation function for all layers except the last, is written as follows:

$$\hat{\phi} = \theta^4 \odot \tanh(\theta^3 \odot \tanh(\theta^2 \odot \tanh(\theta^1 \odot \mathbf{X}))). \quad (17)$$

The model output is considered as an approximate solution for the scalar potential ϕ of PDEs. Given the scalar potential and vector potential, the displacement is solved by leveraging the automatic differentiation (AD) operator G in the following equation:

$$\begin{aligned} \hat{\mathbf{u}} &= (\hat{u}_x, \hat{u}_y)^\top = \left(\frac{\partial \hat{\phi}}{\partial x} + \frac{\partial \hat{\psi}}{\partial y}, \frac{\partial \hat{\phi}}{\partial y} - \frac{\partial \hat{\psi}}{\partial x} \right)^\top \\ &= [G(\hat{\phi}, x) + G(\hat{\psi}, y), G(\hat{\phi}, y) - G(\hat{\psi}, x)]^\top \end{aligned} \quad (18)$$

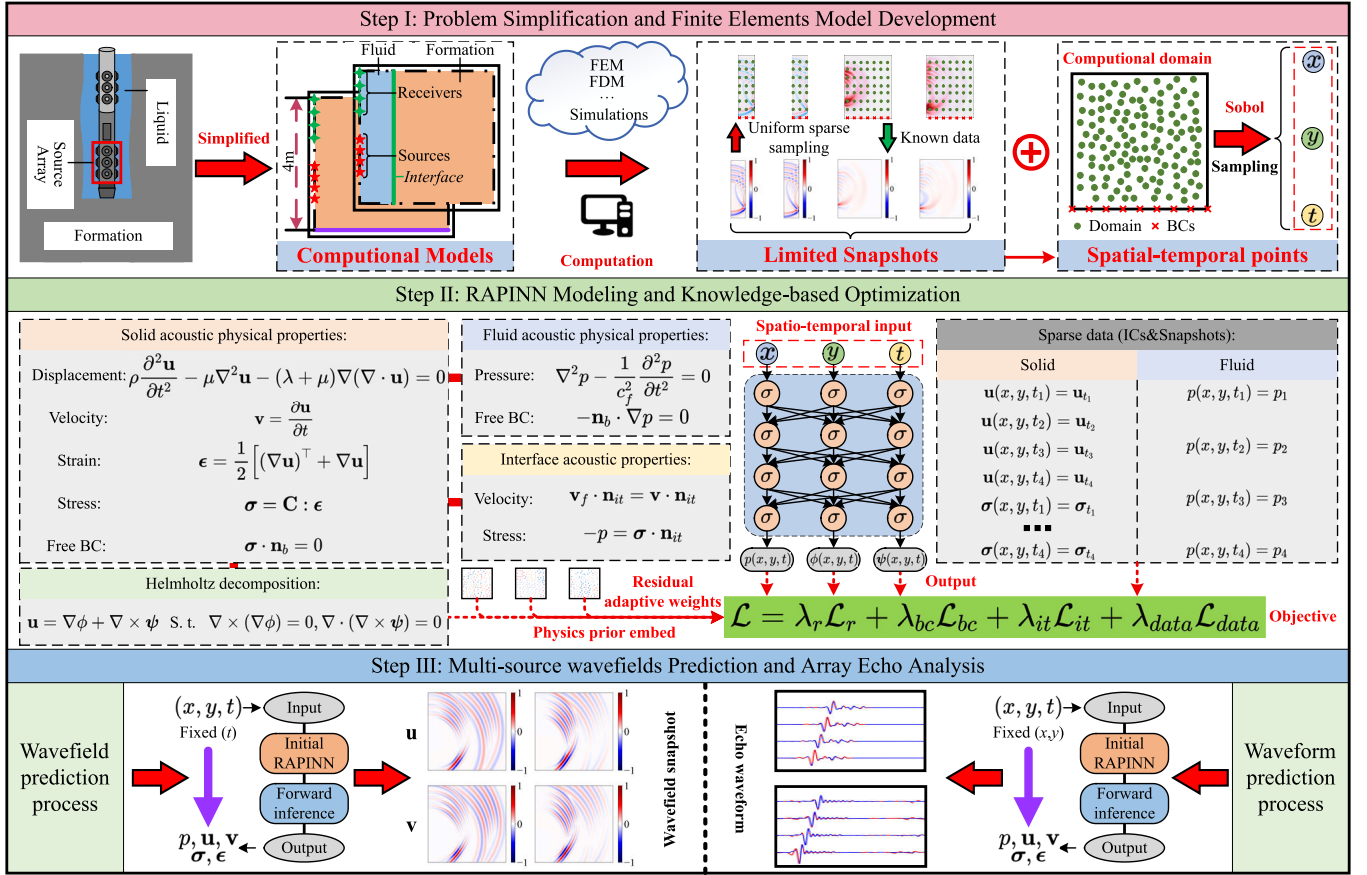


Fig. 3. Schematic of the acoustic phased-array-based forward radiation wavefield modeling method based on RAPINN.

where the $\hat{u}_x$ and $\hat{u}_y$ are the components of predicted $\hat{\mathbf{u}}$. Once the displacement $\hat{\mathbf{u}}$ is obtained, the velocity $\hat{\mathbf{v}}$, strain $\hat{\boldsymbol{\epsilon}}$, and stress $\hat{\boldsymbol{\sigma}}$ in solid domain are derived

$$\begin{aligned} \hat{\mathbf{v}} &= (\hat{v}_x, \hat{v}_y)^T = \left(\frac{\partial \hat{u}_x}{\partial t}, \frac{\partial \hat{u}_y}{\partial t} \right)^T \\ &= [G(\hat{u}_x, t), G(\hat{u}_y, t)]^T \\ \hat{\boldsymbol{\epsilon}} &= (\hat{\epsilon}_{xx}, \hat{\epsilon}_{yy}, \hat{\epsilon}_{xy})^T = \left(\frac{\partial \hat{u}_x}{\partial x}, \frac{\partial \hat{u}_y}{\partial y}, \frac{1}{2} \frac{\partial \hat{u}_x}{\partial y} + \frac{1}{2} \frac{\partial \hat{u}_y}{\partial x} \right)^T \\ &= [G(\hat{u}_x, x), G(\hat{u}_y, y), G(\hat{u}_x, 2y) + G(\hat{u}_y, 2x)]^T \end{aligned} \quad (19)$$

where $\hat{v}_x$ and $\hat{v}_y$ refer to the x - and y -ordinate components of velocity. Similarly, $\hat{\epsilon}_{xx}$, $\hat{\epsilon}_{yy}$, and $\hat{\epsilon}_{xy}$ are the components of strain. To reduce training complexity, stress $\hat{\boldsymbol{\sigma}}$ is derived based on strain [as shown in (3)] rather than displacement. Although the expression differs, the essence remains the same. $\hat{\boldsymbol{\sigma}}$ is defined in the following equation:

$$\begin{aligned} \hat{\sigma}_{xx} &= (\lambda + 2\mu) \hat{\epsilon}_{xx} + \lambda \hat{\epsilon}_{yy} \\ \hat{\sigma}_{yy} &= \lambda \hat{\epsilon}_{xx} + (\lambda + 2\mu) \hat{\epsilon}_{yy} \\ \hat{\sigma}_{xy} &= 2\mu \hat{\epsilon}_{xy}. \end{aligned} \quad (20)$$

Here, the Lamé constants (λ and μ) further constitute the constitutive tensor $\mathbf{C}$ in (3). This method of deriving stress from strain, to some extent, avoids the quadratic computational complexity associated with direct derivation from displacement. Finally, all physical variables and their components in each direction can be solved.

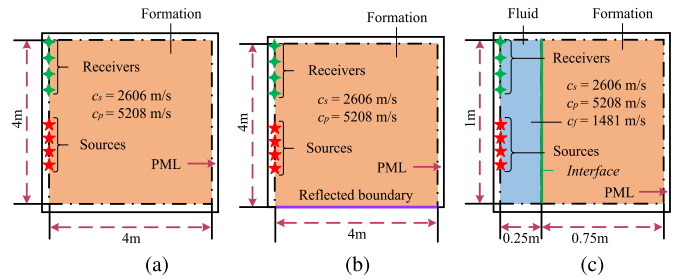


Fig. 4. Schematic of the computational model. (a) Infinite model. (b) Semi-infinite model. (c) Fluid-solid model.

E. Implementation Details and Training Strategy

This section provides a detailed explanation of the computational model, data preprocessing, and training strategies. These strategies ensure that RAPINN accurately converges to precisely predict coupled acoustic wavefields.

1) *Computational Cases*: As shown in Fig. 1, a simplified 2-D elastic medium computational model is solved. Further details are shown in Fig. 4. Fig. 4(a) shows a 4×4 m² granite medium computational domain surrounded by the perfectly matched layers (PMLs). This configuration is used to simulate the propagation of a phased-array wavefield through an infinite formation medium. A free boundary is applied to the bottom of the domain as a semi-infinite domain boundary in Fig. 4(b). This simulates the process where the acoustic wavefield encounters a forward geological boundary

Algorithm 1 Phased-Array-Based Acoustic Wavefield Modeling for Fluid–Solid Structure Using RAPINN (Training Stage)

Input: discrete spatial–temporal coordinates $\mathbf{X} = \{\mathbf{X}_r, \mathbf{X}_{data}, \mathbf{X}_{it}\}$, and known sparse data $\mathbf{U}_{data}$ at coordinates $\mathbf{X}_{data}$.
Output: trained RAPINN.

Initializing RAPINN parameters, weight matrix $\mathbf{W}$ and iteration $e = 0$. Setting epochs $E_1 = 100000$ and $E_2 = 50000$.

- 2 **while** $e < E_1$ **do**
 - Calculate the loss terms $\mathcal{L}_r$ as in Eq. (9) at spatial–temporal coordinates $\mathbf{X}_r$ without any known data.
- 4 Calculate multiple physical variables $\hat{p}, \hat{\mathbf{u}}, \hat{\mathbf{v}}, \hat{\sigma}, \hat{\epsilon}$ using Eqs. (1), (2) and (3). Then, calculate the loss terms $\mathcal{L}_{data}$ as in Eq. (10) at spatial–temporal coordinates $\mathbf{X}_{data}$ with known data $\mathbf{U}_{data}$.
 Calculate multiple physical variables $\hat{p}, \hat{\mathbf{u}}, \hat{\mathbf{v}}, \hat{\sigma}, \hat{\epsilon}$ using Eqs. (1), (2) and (3). Then, calculate the loss terms $\mathcal{L}_{it}$ as in Eq. (9) at spatial–temporal coordinates $\mathbf{X}_{it}$ without any known data.
- 6 Minimizing the loss function $\mathcal{L}$ in Eq. (14) and gradient back propagation.
 Update network parameters and weight matrix $\mathbf{W}$ with Adam Optimizer, $e = e + 1$.
- 8 **end while**
- while** $e < (E_1 + E_2)$ **do**
 - 10 Calculate the loss terms $\mathcal{L}_r$ as in Eq. (9) at spatial–temporal coordinates $\mathbf{X}_r$ without any known data.
 Calculate multiple physical variables $\hat{p}, \hat{\mathbf{u}}, \hat{\mathbf{v}}, \hat{\sigma}, \hat{\epsilon}$ using Eqs. (1), (2) and (3). Then, calculate the loss terms $\mathcal{L}_{data}$ as in Eq. (10) at spatial–temporal coordinates $\mathbf{X}_{data}$ with known data $\mathbf{U}_{data}$.
 - 12 Calculate multiple physical variables $\hat{p}, \hat{\mathbf{u}}, \hat{\mathbf{v}}, \hat{\sigma}, \hat{\epsilon}$ using Eqs. (1), (2) and (3). Then, calculate the loss terms $\mathcal{L}_{it}$ as in Eq. (9) at spatial–temporal coordinates $\mathbf{X}_{it}$ without any known data.
 Minimizing the loss function $\mathcal{L}$ in Eq. (14) and gradient back propagation.
 - 14 Only update network parameters with L-BFGS Optimizer, and the self-adaptive weight matrix $\mathbf{W}$ is fixed, $e = e + 1$.
- end while**
- 16 **Return** trained RAPINN.

and generates reflected waves. Fig. 4(c) uses a fluid–solid structure to simulate the propagation, refraction, and reflection of wavefields in the specific medium composition of the borehole. In all cases, the formation is set to granite, and the fluid medium is set to water. The speed of the P-wave in the solid domain is 5208 m/s, while the speed of the S-wave is 2606 m/s. Meanwhile, the speed of the P-wave in the fluid domain is 1481 m/s. The PMLs surrounding the boundary absorb waves outside the domain to prevent redundant interference.

As also shown in Fig. 4, multiple receivers are placed above the sources to observe both the direct wave in the fluid and the reflected wave from the front boundary. Multiple excitation sources are applied to the left boundary of the computational domain. The acoustic phased-array wavefield is simulated by modulating the phase difference. There are two scenarios encompassed by the number of sources: the four- and eight-source configurations commonly used in engineering. The deflection angles used to synthesize wavefields are 45° and 90° , which simulate typical engineering conditions. The excitation signal uses a sinusoidally modulated Gaussian pulse signal.

2) *Data Preprocessing*: The data normalization is employed to eliminate the effect of different input spatial–temporal coordinates magnitudes. The normalization formula is defined as: $\mathbf{X}_{norm} = (2(\mathbf{X} - \mathbf{X}_{min})) / (\mathbf{X}_{max} - \mathbf{X}_{min}) - 1$, where $\mathbf{X} = \{x, y, t\}$ is the input vector of the sampling point, which is a 3-D data pair in time and space coordinates. The $\mathbf{X}_{max}$ and $\mathbf{X}_{min}$ are the maximum and minimum values of the input data, respectively. $\mathbf{X}_{norm}$ is the normalized input data with a range of $[-1, 1] \in \mathbb{R}$. Meanwhile, the wavefield variables, including p , $\mathbf{u}$, $\mathbf{v}$, σ , and ϵ , are normalized by

their respective maximum values to mitigate convergence difficulties caused by differing magnitudes.

3) *Training Strategy*: All numerical examples are executed on a server system with an Intel¹ Xeon¹ Gold 6248R central processing unit (CPU), an NVIDIA RTX A6000 graphics processing unit (GPU) with 48 GB memory, and 512 GB random access memory. Furthermore, there are 10 201 training points in the domain and on the reflected boundaries using the Sobol sequence sampling method. Meanwhile, 10 201 supervised training points are obtained from each wavefield snapshot by the uniform sampling method, and their resolution is lower than the predicted wavefields at other times. The training process is divided into two stages. In the first stage, the Adam optimizer is used for 100 000 epochs to achieve convergence. The second stage uses the L-BFGS optimizer to train for 50 000 epochs, fine-tuning the results. The initial learning rates are set to $3e^{-3}$ and 1, respectively. Algorithm 1 details the RAPINN training process in the fluid–solid composite structure case.

The standard PINN [18] is selected as the benchmark to illustrate the effectiveness and superiority of the proposed method. All methods use the same amount of sparse data and the same training parameters and network structure. The FEM numerical simulation results are used as benchmark references for the evaluation. In addition, root-mean-square error (RMSE), $\mathcal{L}_2$ -norm error (RE), and coefficient of determination (R^2) are used to evaluate the wavefields prediction quality.

III. NUMERICAL CASES AND DISCUSSIONS

In this section, three computational cases are used to validate the effectiveness of RAPINN. As shown in Fig. 4,

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TABLE I
COMPONENTWISE SOLUTION FOR DIFFERENT WAVEFIELD VARIABLES IN
THE INFINITE-DOMAIN CASE

Variable	Methods	$RMSE$	RE	R^2
u_x	PINN	4.9137×10^{-3}	0.0919	0.9916
	RAPINN	3.5100×10^{-3}	0.0656	0.9957
u_y	PINN	5.1804×10^{-3}	0.0813	0.9934
	RAPINN	4.2479×10^{-3}	0.0667	0.9956
v_x	PINN	5.1778×10^{-3}	0.0960	0.9908
	RAPINN	4.4353×10^{-3}	0.0823	0.9932
v_y	PINN	6.0882×10^{-3}	0.0980	0.9904
	RAPINN	5.2632×10^{-3}	0.0847	0.9928
ϵ_{xx}	PINN	5.9049×10^{-3}	0.0929	0.9914
	RAPINN	5.2088×10^{-3}	0.0820	0.9933
ϵ_{xy}	PINN	3.8166×10^{-3}	0.1197	0.9857
	RAPINN	3.1418×10^{-3}	0.0985	0.9903
ϵ_{yy}	PINN	5.2240×10^{-3}	0.0923	0.9915
	RAPINN	4.5604×10^{-3}	0.0806	0.9935
σ_{xx}	PINN	6.4964×10^{-3}	0.0928	0.9914
	RAPINN	5.7447×10^{-3}	0.0821	0.9928
σ_{xy}	PINN	3.5945×10^{-3}	0.1197	0.9857
	RAPINN	2.9596×10^{-3}	0.0985	0.9903
σ_{yy}	PINN	5.3619×10^{-3}	0.0916	0.9916
	RAPINN	4.6646×10^{-3}	0.0797	0.9936

it includes finite homogeneous granite medium, semi-infinite granite medium, and fluid–solid composite structures.

A. Homogeneous Solid Model

As shown in Fig. 4(a), four sources and four receivers are placed along the left boundary. Each source applies a sinusoidally modulated Gaussian pulse in the x -direction as the excitation signal. Simultaneously, time delays are introduced between the sources to simulate acoustic phased-array wavefields. Specifically, the excitation signal $S(x, t)$ is defined as

$$S(x, t) = \exp\left(-\left(\frac{(x - x_s)^2}{d_s}\right)\right) \exp\left(-\left(\frac{(t - \tau)}{(t_0/2)}\right)^2\right) \times \sin(2\pi f(t - \tau)) \quad (21)$$

where x_s and d_s represent the source's location and areas. $f = 10$ kHz and $t_0 = 1/f$ s represent the modulation signal frequency and period, respectively. $\tau = k0.25 \sin(\pi/2)/c_p$, where $k \in \{0, 1, 2, 3\}$ is the excitation delay time for the k th source. In this case, a total of four sources control the delay time to achieve a 90° deflection angle in the synthesized wavefield. This simulates the propagation process of the wavefield directly in front of the drill bit. Acoustic waves are absorbed by the PMLs outside the computational domain, so no reflected waves are generated. Therefore, the acoustic phased-array wavefield only propagates forward.

The wavefields from the FEM at 0.4 and 0.5 ms are used as ICs, and the subsequent wavefields at 0.8, 1.0, 1.2, and 1.4 ms are used as ground truth to verify the prediction accuracy.

According to (11), there are neither $\mathcal{L}_{bc}$ and $\mathcal{L}_{it}$ terms nor any snapshot data loss ($\mathbf{U}^n$) in $\mathcal{L}_{data}$. Table I lists the componentwise prediction results for different wavefield variables. For u_x , the proposed method achieves the best prediction accuracy, with an R^2 of 0.9957 and a $\mathcal{L}_2$ norm error of 0.0656. For each component, RAPINN maintains high accuracy, with R^2 exceeding 0.99. Compared to the standard PINN, the proposed method demonstrates improvements in multiple quantitative metrics. For ϵ_{xy} , R^2 increased by 0.0046 and the $\mathcal{L}_2$ norm error decreased by 0.0212.

Figs. 5 and 6 show the visualized results of four acoustic sources in an infinite domain. RAPINN accurately describes the forward propagation process of the phased-array acoustic wavefields in an infinite domain with low error. This demonstrates the superior wavefield modeling capability of RAPINN. The velocity difference between P- and S-waves causes these two-mode waves to exhibit spatial–temporal separation during the propagation process. As shown in Figs. 5 and 6 by the red and green boxes, the proposed RAPINN clearly decouples the two-mode waves in each component direction with high modeling accuracy. This validates the effectiveness of the Helmholtz-decomposition-based explicit constraint condition mentioned in Section II-A2, demonstrating its excellent P- and S-waves decoupling capability.

In addition, Fig. 7 shows the u_x and v_x time-domain waveform received by the four receivers. Since no reflected waves are generated in the infinite domain, only one incident wave peak is present. The predicted values from the proposed method show high consistency with the FEM benchmark in their temporal evolution. The peak arrival time is delayed as the distance between the receiver and the acoustic source increases. RAPINN accurately simulates this phenomenon. This demonstrates the excellent time-tracking capability of the proposed method. Moreover, the hyperparameters are also listed in Appendix.

B. Reflected Boundary Solid Model

As shown in Fig. 4(b), the semi-finite domain computational model consists of eight acoustic sources and receivers placed along the left boundary. The same excitation signal is used as in the infinite-domain computational model, but the source distance is reduced to half of its original value. In this case, eight acoustic sources control delay times are used to create a 45° deflection angle within the synthetic wavefields. A reflection boundary is set at the bottom to simulate the reflected wavefield generated by the geological boundary in front of the drill bit. This computational model simulates the forward propagation and reflection of the phased-array acoustic wavefields. In addition, it simulates the mode conversion process between P- and S-waves caused by the free boundary.

The 0.4- and 0.5-ms wavefield snapshots from the FEM are used as ICs, and the 0.6, 0.8, 1.0, and 1.2 ms subsequent wavefields are used as ground truth. Considering the complexity of mode conversion at boundaries, two intermediate wavefield snapshots (with only displacement and stress) are introduced as known data. Supervised learning with sparse

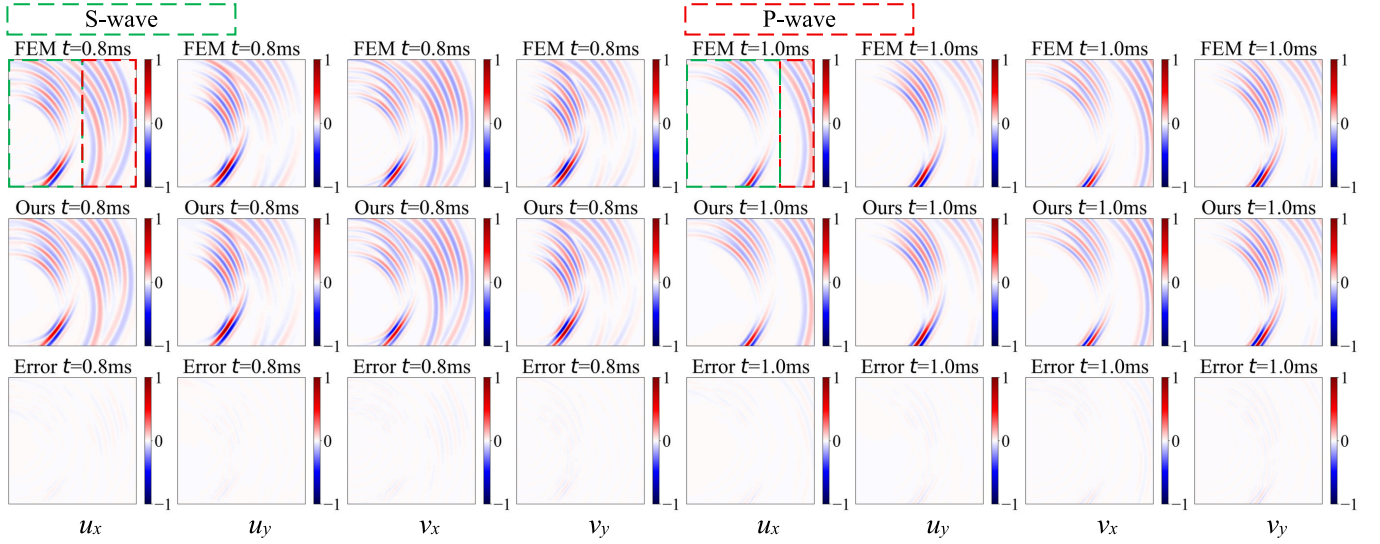


Fig. 5. Componentwise comparison of displacement and velocity in the infinite domain.

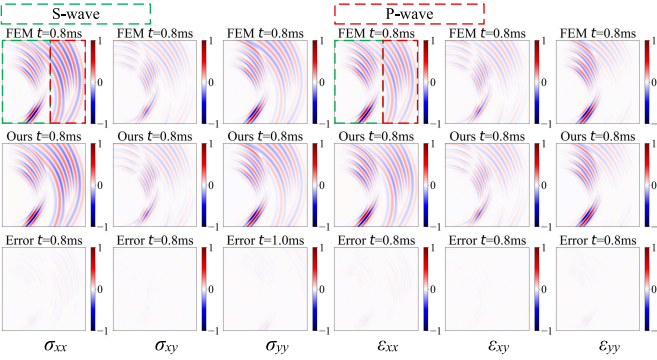


Fig. 6. Componentwise comparison of stress and strain in the infinite domain.

data improves waveform reconstruction quality, thereby alleviating this challenging mode-conversion problem. Therefore, according to (11), the BCs loss terms $\mathcal{L}_{bc}$ and snapshot data loss ($\mathcal{L}_{sn}$) in $\mathcal{L}_{data}$ are all nonzero. These loss terms constrain the neural network to generate mode conversions of various waves at the boundary. These conversions result in a more spatial-temporally overlapping reflected wavefield. By imposing boundary constraints and utilizing limited known data, the learning capability of the neural network for reflection wave mode conversion and overlapping acoustic wavefields is enhanced. Table II also lists the componentwise prediction results for different variables. For all components, the proposed RAPINN also maintains high solution accuracy, with R^2 exceeding 0.98 and $\mathcal{L}_2$ norm error both below 0.12. By contrast, the standard PINN lacks optimization for adaptive weights and therefore has limitations when it comes to solving for the individual components of displacement and velocity. For instance, when solving for the u_x component, PINN achieves an R^2 of only 0.9369, while the RAPINN achieves an R^2 of 0.9896 and an improvement of 0.0527. This improvement is crucial for reconstructing wavefield details. Furthermore, the proposed RAPINN achieves accurate componentwise solutions for multiple acoustic physical variables—including displacement, velocity, stress, and

TABLE II
COMPONENTWISE SOLUTION FOR DIFFERENT WAVEFIELD VARIABLES IN THE SEMI-INFINITE-DOMAIN CASE

Variable	Methods	$RMSE$	RE	R^2
u_x	PINN	2.1513×10^{-2}	0.2513	0.9369
	RAPINN	8.7515×10^{-3}	0.1022	0.9896
u_y	PINN	1.9725×10^{-2}	0.2536	0.9357
	RAPINN	6.9818×10^{-3}	0.0897	0.9920
v_x	PINN	2.0457×10^{-2}	0.2410	0.9419
	RAPINN	9.5822×10^{-3}	0.1129	0.9873
v_y	PINN	2.1141×10^{-2}	0.2717	0.9262
	RAPINN	7.1849×10^{-3}	0.0924	0.9915
ϵ_{xx}	PINN	2.0818×10^{-2}	0.2753	0.9242
	RAPINN	8.1095×10^{-3}	0.1073	0.9885
ϵ_{xy}	PINN	1.4557×10^{-2}	0.2509	0.9370
	RAPINN	6.4704×10^{-3}	0.1153	0.9876
ϵ_{yy}	PINN	2.0202×10^{-2}	0.2832	0.9198
	RAPINN	7.8576×10^{-3}	0.1101	0.9879
σ_{xx}	PINN	2.1431×10^{-2}	0.2477	0.9387
	RAPINN	9.1809×10^{-3}	0.1061	0.9887
σ_{xy}	PINN	1.3311×10^{-2}	0.2509	0.9371
	RAPINN	5.9177×10^{-3}	0.115	0.9876
σ_{yy}	PINN	2.0421×10^{-2}	0.2545	0.9352
	RAPINN	8.8067×10^{-3}	0.1098	0.9880

strain—in a semi-infinite solid medium. This demonstrates the robust capabilities of the componentwise solutions.

Figs. 8 and 9 show the visualized results of eight acoustic sources in a semi-infinite solid domain. RAPINN accurately describes the forward propagation and reflected wave propagation processes of phased-array acoustic wavefields in semi-infinite domains with low error. Particularly in regions near the free boundary, the mode conversion between P- and S-waves prevents the neural network from accurately learning the evolution of each mode wave. However, the RAPINN

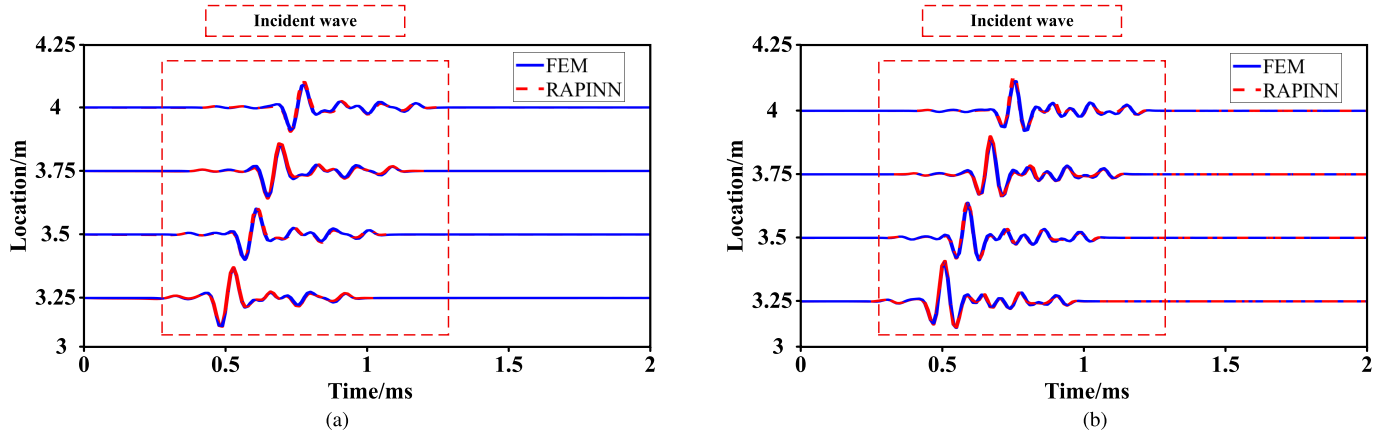


Fig. 7. Comparison of time history at different receiver locations in the infinite case. (a) u_x waveform varying with time. (b) v_x waveform varying with time.

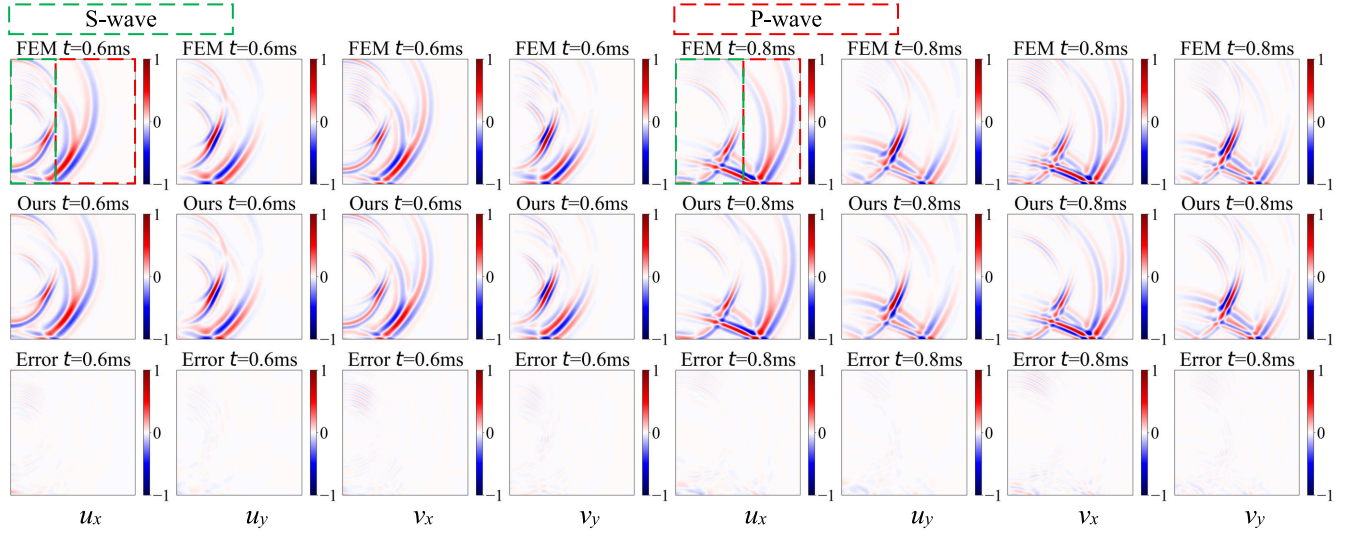


Fig. 8. Componentwise comparison of displacement and velocity in the semi-infinite domain.

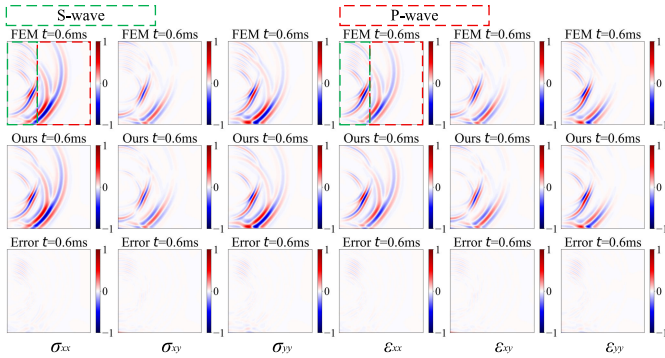


Fig. 9. Componentwise comparison of stress and strain in the semi-infinite domain.

successfully addresses this issue by using the residual adaptive weights assignment mechanism and Helmholtz decomposition. It achieves minimal error near the boundary and enables precise, componentwise acoustic wavefield reconstruction. In summary, the proposed RAPINN demonstrates its outstanding P- and S-waves decoupling capability by decoupling the two mode waves with high precision in each component direction.

Furthermore, Fig. 10 also shows the u_x and v_x time-domain waveform received by the four receivers. Due to the reflected waves generated by the bottom reflected boundary, two peak regions appear in the received waveform. Before the first direct wave arrives, the nonzero wavefront prediction errors are observed in RAPINN. This is primarily attributed to its noncausal characteristics. Specifically, under the global optimization strategy, the neural network sacrifices local causal constraints to satisfy strong boundary constraints. Consequently, wavefront energy undergoes numerical diffusion and results in energy leakage, which leads to prediction errors. Although there exist weak amplitude fluctuations before the direct wave arrives, its main peak and subsequent reflected waves show excellent consistency with the FEM reference results, confirming that RAPINN effectively captures the evolution of the weak reflected waves. It is worth noting that the phenomenon of weak echo signals exhibits high consistency in actual logging, as energy dispersion and other factors result in extremely weak echo signals. RAPINN fundamentally reconstructs the amplitude characteristics of echo signals and their time-evolution process, demonstrating its exceptional temporal tracking capability.

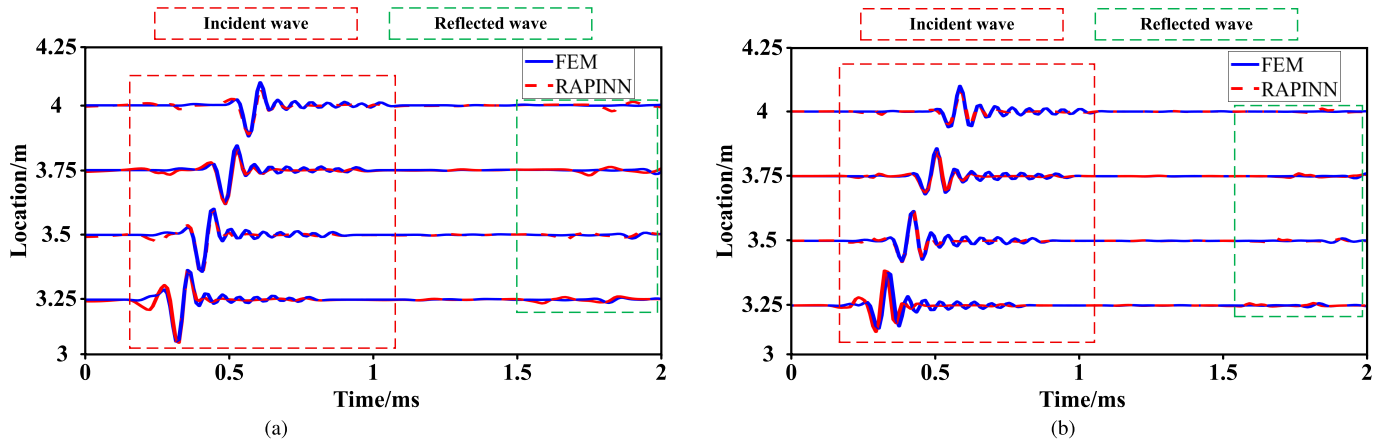


Fig. 10. Comparison of time history at different receiver locations in the semi-infinite case. (a) u_x waveform varying with time. (b) v_x waveform varying with time.

C. Fluid–Solid Composite Model

To simulate the propagation and reflection behavior of phased-array acoustic wavefields within borehole structures more accurately, an acoustic wavefield surrogate model based on RAPINN is developed. As shown in Fig. 4(c), the semi-structure of the borehole is simulated using a computational model consisting of a fluid–solid dual-layer medium, with the fluid medium on the left and the solid medium on the right. Four acoustic sources and four receivers are positioned along the left boundary. Each excitation source signal is a sinusoidally modulated Gaussian pulse, which enables the simulation of a phased-array acoustic wavefield by modulating the delay time of the different excitation sources. In this case, four acoustic source control delay times are used to create a 45° deflection angle within the synthetic wavefield. The excitation signals are defined in the following equation:

$$S(t) = 100 \exp\left(-\left(\frac{(t-\tau)}{(t_0/2)}\right)^2\right) \sin(2\pi f(t-\tau)). \quad (22)$$

Since the fluid medium cannot generate or propagate S-waves, the acoustic wavefield in the fluid–solid structure becomes more complex. It primarily involves two processes: first, the excitation signal generates a pure P-wave wavefield in the fluid medium, which then propagates forward. When these waves reach the solid interface, some of them reflect due to impedance differences, generating reflected P-waves that overlap with the fluid wavefield. The others enter the solid medium, undergo mode conversion, and generate a P- and S-waves-coupled acoustic wavefield.

As shown in Fig. 3, RAPINN outputs $\hat{p}$, $\hat{\phi}$, and $\hat{\psi}$. By leveraging interface physical constraints, it couples the fluid and solid wavefields simultaneously to establish a unified acoustic wavefield surrogate model. The 0.28 and 0.30 ms wavefields from the FEM are used as ICs, and the 0.32, 0.34, 0.36, and 0.38 ms subsequent wavefields are used as ground truth. Meanwhile, two intermediate wavefield snapshots are introduced as known data. Therefore, the loss terms $\mathcal{L}_{it}$, $\mathcal{L}_{ic}$ terms, and snapshot data loss ($\mathcal{L}_{sn}$) in $\mathcal{L}_{data}$ are all nonzero. These loss terms constrain the proposed RAPINN to generate mode conversions of various waves at the interface. By imposing interface constraints and utilizing limited known data,

TABLE III
COMPONENTWISE SOLUTION FOR DIFFERENT WAVEFIELD VARIABLES IN THE FLUID–SOLID COMPLEX CASE

Variable	Methods	$RMSE$	RE	R^2
p	PINN	3.4444×10^{-3}	0.3386	0.8847
	RAPINN	3.2042×10^{-3}	0.3150	0.9003
u_x	PINN	7.5409×10^{-3}	0.6901	0.9916
	RAPINN	8.3368×10^{-3}	0.0763	0.9897
u_y	PINN	6.8221×10^{-3}	0.1836	0.9662
	RAPINN	7.0376×10^{-3}	0.1894	0.9641
v_x	PINN	1.324×10^{-3}	0.1633	0.9729
	RAPINN	1.2185×10^{-2}	0.1757	0.9686
v_y	PINN	1.0140×10^{-2}	0.3444	0.8813
	RAPINN	1.0101×10^{-2}	0.3431	0.8822
ϵ_{xx}	PINN	8.3979×10^{-3}	0.2293	0.9453
	RAPINN	9.0096×10^{-3}	0.2460	0.9370
ϵ_{xy}	PINN	8.0551×10^{-3}	0.2156	0.9535
	RAPINN	7.3501×10^{-3}	0.1967	0.9613
ϵ_{yy}	PINN	8.9332×10^{-3}	0.2589	0.9328
	RAPINN	9.4640×10^{-3}	0.2743	0.9246
σ_{xx}	PINN	7.8454×10^{-3}	0.2577	0.9304
	RAPINN	7.8441×10^{-3}	0.2577	0.9304
σ_{xy}	PINN	5.3318×10^{-3}	0.2156	0.9535
	RAPINN	4.8650×10^{-3}	0.1967	0.9613
σ_{yy}	PINN	8.3476×10^{-3}	0.2960	0.9123
	RAPINN	8.2999×10^{-3}	0.2943	0.9133

the learning capability of the neural network for reflection wave mode conversion and overlapping acoustic wavefields is enhanced. Table III also lists the pressure and componentwise wavefield prediction results. Especially, for pressure in fluid domains, the proposed RAPINN maintains high accuracy, with an R^2 exceeding 0.90. Compared to the standard PINN, RAPINN achieves higher accuracy in pressure solutions, with an R^2 improvement of 0.0156 and a relative error reduction of 0.0236, enabling superior perception of wavefield details. Furthermore, the proposed RAPINN achieves accurate

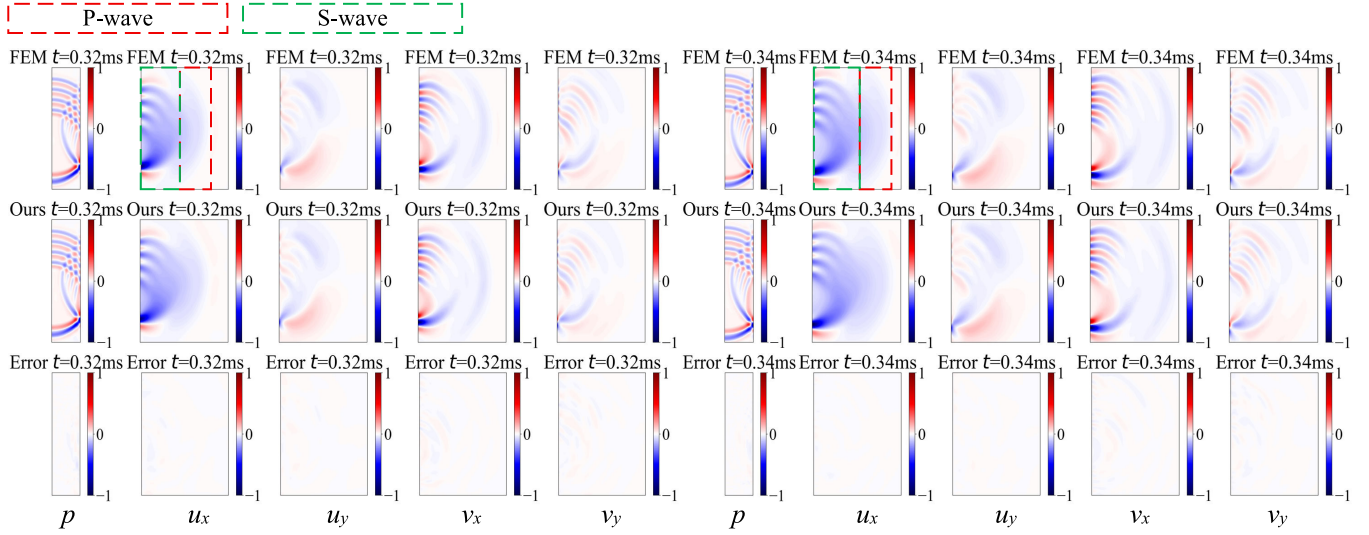


Fig. 11. Componentwise comparison of pressure, displacement, and velocity in the fluid-solid composite structures.

componentwise solutions for multiple variables—including displacement, velocity, stress, and strain—in a fluid-solid complex model. The R^2 for all component solutions is greater than 0.88, which demonstrates the robust capabilities of the componentwise solutions.

Figs. 11 and 12 show the visualized results of four acoustic sources in a fluid-solid complex domain. Particularly in regions near the interface, under conditions of limited data supervision, RAPINN successfully learned to distinguish between fluid acoustic wavefields and solid acoustic wavefields. It achieves minimal error near the interface and enables precise, componentwise acoustic wavefield reconstruction. Meanwhile, the visualization results reveal that P-waves in the fluid undergo mode conversion at the interface, subsequently exciting a composite acoustic wavefield with P- and S-waves overlapping within the solid medium. RAPINN has successfully learned this characteristic, enabling the precise prediction of the directional components of multiple acoustic wavefield variables within the solid medium. In summary, RAPINN is a feasible tool for decoupling P- and S-waves in fluid-solid composite wellbore structures. It achieves accurate simulation of the acoustic wavefield within the wellbore structure by decoupling the two-mode waves in each component direction.

D. Discussions

Based on the above comprehensive analysis, the proposed RAPINN effectively simulates the propagation, refraction, and reflection phenomena of phased-array synthetic acoustic waves under multisource, multiboundary, and multiple deflection angle conditions. It also demonstrates robust modeling capabilities for acoustic wavefields in fluid-solid composite structures. This section provides a detailed analysis of the ablation experiment results and the limitations of the proposed method.

1) *Ablation Study on RA Weight Assignment Mechanism:* First, the only difference between the standard PINN and the proposed RAPINN is the absence of the RA pointwise weights assignment mechanism. Experiments comparing the

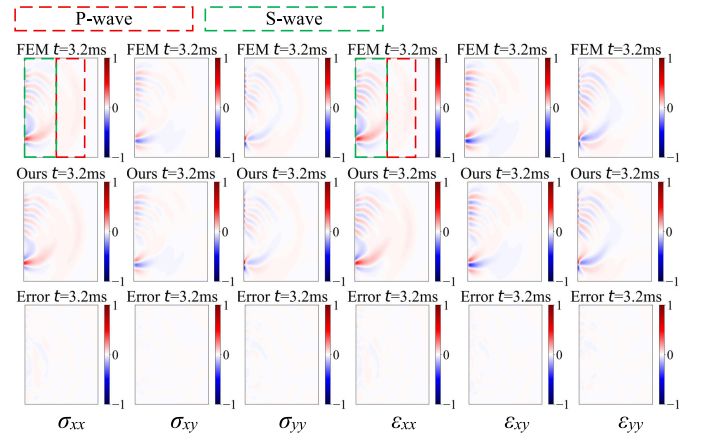


Fig. 12. Componentwise comparison of stress and strain in the fluid-solid composite structures.

two methods can be considered as an ablation study of the RA mechanism. Based on the experimental results in Sections III-A–III-C, the PINN-based phased-array wavefield solving method exhibits superior modeling capabilities to standard PINN in a variety of acoustic wavefield scenarios. This adaptive mechanism enables PINN to capture wavefield details more effectively, thereby enhancing the accuracy of wavefield simulations. Particularly in semi-infinite reflection wavefields, the fixed-weight approach struggles to adapt to complex reflection wavefields, whereas RAPINN solves overlapping reflection wavefields with high accuracy. Moreover, Fig. 13 shows the pointwise weight visualization results for PINN and RAPINN in the semi-infinite case. The RA weights assign mechanism modifies the traditional macrolevel weight distribution approach by enabling adaptive weight updates based on residuals and their corresponding gradients. The difference between $\mathbf{W}_2$ and $\mathbf{W}_3$ further demonstrates that different governing equations require different weights for optimization, rather than fixed weights. Meanwhile, the weights assigned to different sampling points are not entirely

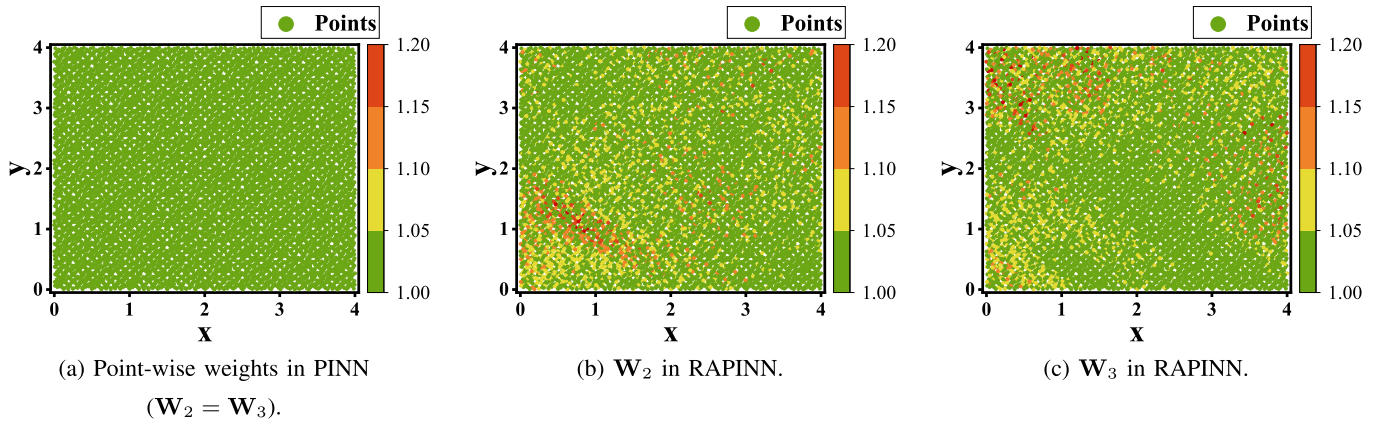


Fig. 13. Comparison of pointwise weight mechanisms. (a) PINN. (b) and (c) RAPINN.

TABLE IV
COMPARISON OF TRAINING TIME IN SEMI-INFINITE CASE

Methods	Platform	Time/s
PINN	GPU: NVIDIA RTX A6000	55863
RAPINN	GPU: NVIDIA RTX A6000	53540

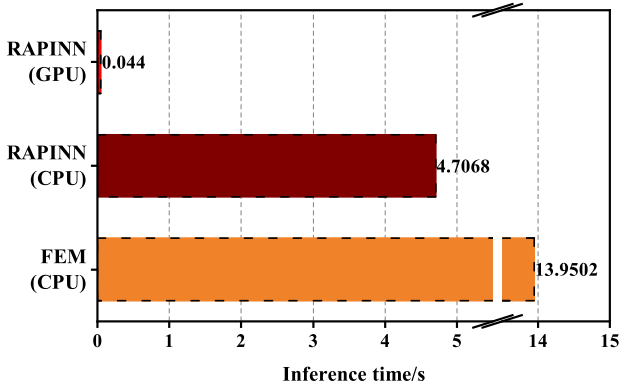


Fig. 14. Comparison of single snapshot inference time in semi-infinite case.

equal to 1, which also demonstrates that the RAPINN is not degenerated to the standard PINN.

2) *Comparison of Training and Inference Costs:* Table IV lists the computational time required to solve semi-infinite domain wavefields on different devices. Under GPU acceleration, standard PINN takes 55863 s to obtain a trained surrogate model. Furthermore, the proposed RAPINN takes 53540 s to obtain a fully trained surrogate model. Compared to PINN, RAPINN reduces training time by 2323 s. RAPINN more rapidly identifies optimal descent directions through pointwise weight optimization, thereby reducing training costs to some extent.

Finally, Fig. 14 shows the wavefield inference time for FEM and RAPINN on different devices. The FEM takes 13.9502 s to obtain a wavefield snapshot on the CPU. RAPINN takes 4.7068 s to obtain a predicted wavefield snapshot in the inference stage. RAPINN only achieves over 66% faster wavefield prediction on the CPU compared to traditional FEM. This is because computing physical variables, such as displacement, velocity, stress, and strain, requires repeated

gradient calculations. Since the CPU is highly sensitive to matrix operations, this results in longer computation times. In addition, RAPINN achieves a single wavefield snapshot in just 0.0440 s with GPU acceleration, which is over $300\times$ faster than the FEM. This significantly enhances the efficiency of wavefield reconstruction. Considering accuracy simultaneously, RAPINN achieves excellent prediction accuracy for multiple wavefield parameters. Specifically, for displacement vectors, R^2 of RAPINN exceeds 0.9890, indicating high consistency with the FEM benchmark. For the v_x component with the largest prediction errors, R^2 also exceeds 0.9870, and the $\mathcal{L}_2$ norm error is less than 0.12. In addition, the CPU-based inference process is faster than the FEM. In other words, RAPINN effectively improves computational efficiency while achieving a similar level of computational accuracy to FEM. This cost analysis reveals that RAPINN significantly improves the efficiency of wavefield prediction in the inference stage after training.

On the other hand, one significant limitation that should not be ignored is the training costs. PINN-like methods require repeated gradient computations, resulting in high training costs. Generally, FEM is highly dependent on meshing strategies and time-stepping strategies. As the mesh is refined and the solution time increases, the computational cost typically grows rapidly. Furthermore, poor mesh design often leads to a significant loss of computational accuracy. In contrast, the PINN-based solution framework is essentially a mesh-free approach that is decoupled from time-stepping constraints. Although the training cost is high, it overcomes the rigid time-step constraints of traditional solvers. Once trained, the network serves as a continuous surrogate, providing instantaneous wavefield predictions at any arbitrary time step. In addition, the parallel computing potential of PINN on GPUs has not yet been fully exploited. Thus, integrating appropriate acceleration techniques further improves solution efficiency [5].

IV. CONCLUSION

In this study, a novel architecture, termed RAPINN, is proposed for modeling the forward radiation wavefields in acoustic phased-array logging scenarios. Based on this framework, three key challenges encountered when solving the 2-D acoustic wavefield using deep learning technologies

TABLE V
NETWORK ARCHITECTURE AND WEIGHTS
CONFIGURATION HYPERPARAMETERS

Models	Network [Input]+[width]×depth+[output]	Weights $\{\lambda_r, \lambda_{bc}, (\lambda_{data}), \lambda_{it}\}$
Infinite	$[3] + [25] \times 5 + [2]$	$\{1, 0, (1,0), 0\}$
Semi-infinite	$[3] + [25] \times 5 + [2]$	$\{1, 1, (1,8), 0\}$
Fluid-solid	$[3] + [25] \times 5 + [3]$	$\{1, 1, (1,10), 1\}$

are addressed. First, a hybrid-driven framework, combining physical priors and sparse data constraints, is proposed. It embeds the governing equations, describing acoustic wavefield propagation characteristics, into the loss function, guiding the training process jointly with sparse data to enhance interpretability and wavefield prediction quality. This significantly reduces the required training wavefield datasets. Second, the Helmholtz decomposition method is employed to decompose the elastic wave equation without explicit constraints into potential-based wave equations with independent constraints for P- and S-waves. This enables the DNN to independently learn the spatial-temporal characteristics of P- and S-waves, achieving automatic decoupling. Third, the RA pointwise weight assignment strategy automatically assigns weights at each point based on residual loss gradients. This enables the neural network to learn more detailed wavefield features. Meanwhile, this adaptive mechanism also reduces training time and accelerates network convergence to some extent. Furthermore, based on these contributions, we first develop a phased-array acoustic wavefield surrogate model for fluid–solid composite structures, enabling the simultaneous solution of fluid–solid coupled acoustic wavefields. Compared with the previous FEM, this meshless technique of RAPINN not only improves the computational efficiency but also significantly reduces the required training data compared to purely data-driven methods. The comprehensive analysis demonstrates that the proposed method has excellent modeling capability for phased-array synthetic wavefield with multiple sources, multiple BCs, and multiple deflection angles, and thus has certain feasibility for wavefield modeling. This provides a foundation for acoustic inversion imaging in phased-array logging, improving our understanding of the propagation mechanisms of phased-array acoustics in logging scenarios.

On the other hand, this method has certain limitations. First, the 2-D plane strain assumption theory used in the work inherently ignores azimuth information. Thus, RAPINN fails to capture the spatial-temporal evolution details of the coupled wavefield in complex 3-D structures. Meanwhile, this artificial imposition of structural deformation mechanisms leads to certain limitations in the dynamic response analysis of complex structures. Second, as mentioned earlier, the training costs are relatively high. This limits the flexibility of PINN-like algorithms to some extent. Nevertheless, RAPINN establishes an efficient and feasible solution framework under computationally constrained conditions. It not only offers extremely fast inference speeds but also eliminates the reliance on meshes inherent in traditional numerical methods, enabling seamless expansion to complex 3-D geometric scenarios.

In future research, we plan to integrate Fourier feature encoding into RAPINN to enhance its expressive capability for high-frequency wavefields and improve prediction accuracy [43], [44]. In addition, we will incorporate transfer learning to adapt modeling capabilities across different media while reducing training time [11], [45].

APPENDIX

Table V lists the network architecture and weight configuration hyperparameters for different loss terms for all cases.

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